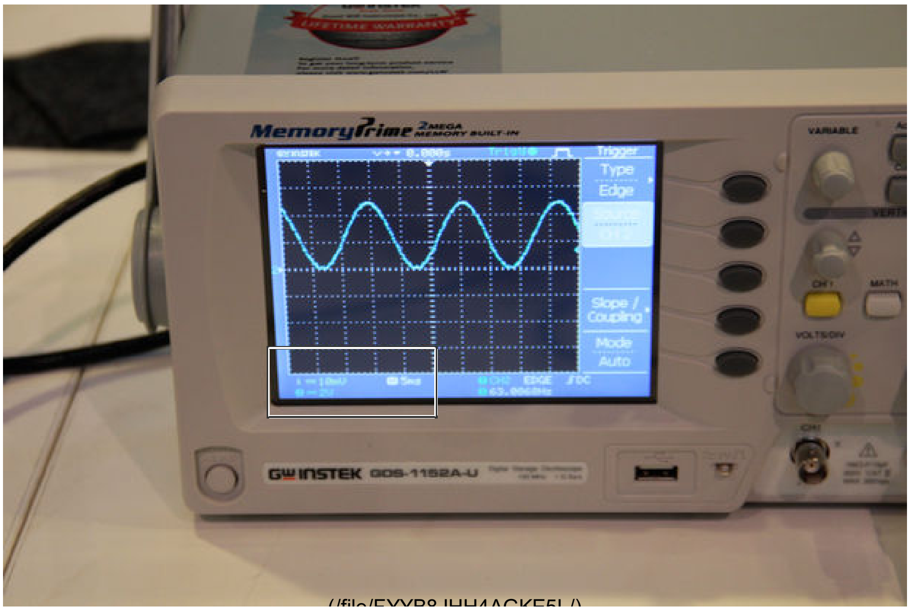
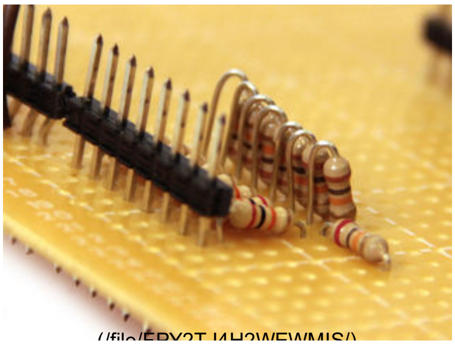
Arduino Timer Interrupts by

amandaghassaei (/member/amandaghassaei/) in arduino (/tag/type-id/category-technology/channel-arduino/)

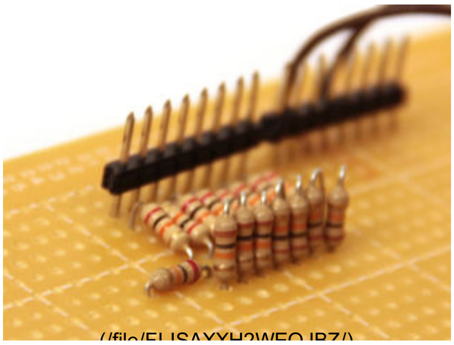
Step 5: Example 3: DAC



(/file/FXYP8 IHH4ACKE5L/)



(/file/FPY2T I4H2WEWMIS/)



(/file/FLJSAXXH2WEO IPZ/)

In this project I used a timer interrupt to output a sine wave of a specific frequency from the Arduino. I soldered a simple 8 bit R2R DAC (http://en.wikipedia.org/wiki/Resistor_ladder) to digital pins 0-7. This DAC was constructed from 10k and 20k resistors arranged in a multi-leveled voltage divider (http://en.wikipedia.org/wiki/Voltage_divider). I'll be posting more about the construction of the DAC in another instructable, for now I've included the photo's above.

I connected the output from the DAC up to an oscilloscope. If you need help understanding how to use/read the oscilloscope check out this tutorial (https://www.instructables.com/id/Oscilloscope-How-To/). I loaded the following code onto the Arduino:

```
//63Hz sine wave
//by Amanda Ghassaei 2012
//https://www.instructables.com/id/Arduino-Timer-Interrupts/

/*
* This program is free software; you can redistribute it and/or modify
* it under the terms of the GNU General Public License as published by
* the Free Software Foundation; either version 3 of the License, or
* (at your option) any later version.
*
*/

//sends 63Hz sine wave to arduino PORTD DAC
float t = 0;
```

I set up a timer interrupt that increments the variable t at a frequency of 40kHz. Once t reaches 627 it resets back to zero (this happens with a frequency of 40,000/628 = 63Hz). Meanwhile, in the main loop the Arduino sends a value between 0 (00000000 in binary) and 255 (11111111 in binary) to digital pins 0 through 7 (PORTD (http://www.arduino.cc/en/Reference/PortManipulation)). It calculates this value with the following equation:

PORTD=byte(127+127*sin(t/100));

So as t increments from 0 to 627 the sine function moves through one complete cycle. The value sent to PORTD is a sine wave with frequency 63Hz and amplitude 127, oscillating around 127. When this is sent through the 8 bit resistor ladder DAC it outputs an oscillating signal around 2.5V with an amplitude of 2.5V and frequency of 63Hz.

The frequency of the sine wave can be doubled by multiplying the (t/100) term by 2, quadrupled by multiplying by 4, and so on...
Also notice that if you increase the frequency of the timer interrupt too much by decreasing the prescaler or OCR2A the sine wave will not output correctly. This is because the sin() function is computationally expensive, and at high interrupt frequencies it does not have enough time to execute. If you are using high frequency interrupts, instead of performing a computation during the interrupt routine, considering storing values in an array and simply calling these values using some kind of index. You can find an example of that in my arduino waveform generator (https://www.instructables.com/id/Arduino-Waveform-Generator/)- by storing 20,000 values of sin in an array, I was able to output sine waves with a sampling frequency of 100kHz.

AravindP4 (/member/AravindP4) 8 days ago Reply

thank u sir

andreondra (/member/andreondra) 5 months ago Reply

Great instructable! Thank you very much!

MalachiB1 (/member/MalachiB1) a year ago Reply

Thank you for breaking this down. I've read a lot of articles about timers but only yours makes it clear!

dineshc31 (/member/dineshc31) ▸ MalachiB1 (/member/MalachiB1) Reply 6 months ago

how exactly this is working for you all ?

For me while compiling in arduino 1.6.10 it is showing

error: 'TIMSK1' was not declared in this scope which is same for all above registers. do we need any #include header file here?

KennyL15 (/member/KennyL15) 9 months ago Reply

Should also mention that the clock prescaler bit names CSX0, CSX1, CSX2 has the first number as 'X'. That number 'X' is linked to the timer number. So if 'X' is 0, then CS00, CS01, CS02 are prescaler bit names for timer 0; and the second value represents the bit position. So C00 means bit 0 (right-most-bit) of the 8-bit register TCCR0B, which is for timer 0.

The reason why I'm indicating this is because the tables (diagram) that go with "prescalers and compare match register" is confusing - since there's no mention about what CS10, CS11, CS12 etc are actually related to.

KennyL15 (/member/KennyL15) 9 months ago Reply

"so a counter will take 10/16,000,000 seconds to reach a value of 9 (counters are 0 indexed)"

Do you actually mean it takes 10/16000000 seconds to begin (initial state) from index 0 and get back to a 0 index again? That is, initial index of 0. Then first increment is 1, second increment is 2, third increment is 3, all the way to 9th increment to index 9, and then one extra increment to get back to index 0 again. Altogether 10 increments to 'get back to 0'.

PeterMortensen (/member/PeterMortensen) 11 months ago Reply

The comment for the line containing "ISR(TIMER2_COMPA_vect)" is confusing. "***timer1***" should probably be "***timer2***".

Sharmila07 (/member/Sharmila07) a year ago Reply

my signal is in the frequency range between 20Hz and 500Hz, so i am taking the sampling frequency as 5000Hz (sampling time will be .2ms), after sampling i need to store the sampled values in the SD card and so i need to use spi. i want to know how much time will spi take to store in SD card and will i be able to sample my signal at this rate? kindly help

---

ParthJ5 (/member/ParthJ5) a year ago Reply

Can anyone please explain the connections to arduino in example 1 ?
I want to know how is the signal generator and Oscilloscope is connected to the Arduino ?

Thanks

★ 

Bruce McCarthy (/member/Bruce McCarthy) a year ago Reply

Brilliant! I struggled with the datasheets for a few days before finding this instructable, and now it's all clear.

Thanks for sharing.

avionics2 (/member/avionics2) a year ago Reply

This is the best interrupts explanations I have come across.

Thanks !

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jerrychee94 (/member/jerrychee94) a year ago Reply

Hi, currently im doing project where it prompt user in serial communication and doing interrupt ISR at the same time. So , when user enter a specific value , it will stop the interrupt for few seconds and then restart the interrupt again. However , when restart the interrupt , the timing is out.(expected interrupt time : 5 microsec per interrupt , actual interrupt time : higher than 5 microsec)

my code is here..

```
void setup()
{
cli();
TCCR0A = 0;
TCCR0B = 0;
TCNT0 = 0;
OCR0A = 80;// = (16*10^6) / (200000*1) - 1 (must be <256)
//200000hz=5microsec
TCCR0A |= (1 << WGM01);
TCCR0B |= (0 << CS02)| (0 << CS01) | (1 << CS00);
TIMSK0 |= (1 << OCIE0A);
sei();
}
void loop()
{
if(serial.available>0)
{
//if user press something...
TIMSK0 &= ~(1 << OCIE0A);
delay(10000);
TIMSK0 |= (1 << OCIE0A);
}
}
```

ISR(TIMER0_COMPA_vect)

{

//code

}

PiraC (/member/PiraC) a year ago Reply

Good Work. Thank you!

sam.gwilliam (/member/sam.gwilliam) a year ago Reply

```
// turn on CTC mode
TCCR0A |= (1 << WGM01);
// Set CS01 and CS00 bits for 64 prescaler
TCCR0B |= (1 << CS01) | (1 << CS00);
```

What is the TCCR0A register? TCCR0B must be the 3-bit one as illustrated (where you set the prescaler) but I don't recall any mention of the other one.

MichealM1 (/member/MichealM1) a year ago Reply

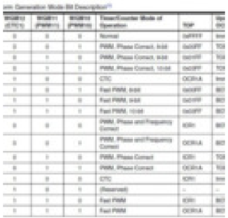
This made it much clearer than other reads I have been to. Based on this explanation, I was able to devise a routine to discipline my software clock based on the 16MHz crystal. One of my units (Uno) is ticking at a very wrong speed. Gaining about 30 seconds per day. That's a LOT! But the software can now take care of it and any other errant crystal it encounters. Thanks!!

MichealM1 (/member/MichealM1) ▸ MichealM1 (/member/MichealM1) Reply a year ago

Sorry. Left out that I am getting NTP time hacks once per day but losing 30 seconds per day is way overboard. Maybe I will replace the crystal and see but as long as I can correct for it, it's not so bad.

Baguda (/member/Baguda) made it! a year ago Reply

In the code you set the timer1 register to CTC mode by "TCCR1B |= (1 << WGM12);". However, for all of the examples using timer0 and timer2, you set the CTC mode by "TCCR0B |= (1 << WGM01);" why do you set bit1 high instead of bit2 for timers 0 and 2 to put them into CTC mode? I found this table online that lists the mode settings(see attached), and it states that (100) is CTC mode. However, it is only for timer1... Is it different for timers 0 and 2?



(https://cdn.instructables.com/FK7/FFSP/IJIUVOFI/FK7FFSPIJIUVOFI.LARGE.jpg)

AnthonyP7 (/member/AnthonyP7) 2 years ago Reply

If the Arduino is executing some code within an interrupt does the counter continue to count up so that the next interrupt is correctly timed?

sandro2204 (/member/sandro2204) ▸ AnthonyP7 (/member/AnthonyP7) Reply a year ago

Yes it does. But you should keep your interupt service routine (ISR) as short as possible. Store a variable, set a flag and that's it. Do time consuming stuff in your main loop.

hal4512 (/member/hal4512) a year ago Reply

according to datasheet of atmega328p p106 -108 when we choose values of COM0A0 and COM0BO for timer0

we set the behavior of pins OC0A OCOB After having config them as output and hardware will proceed as its describe in the datasheet.

all modes Normal mode ,CTC mode non PWM ..ect ..are described in page 100

when you'll read:For generating a waveform output in CTC mode the OC0A output can be set to toggle each logical level on each compare match (you don't have to toggle it by soft).

hal4512 (/member/hal4512) a year ago Reply

you don' need to write a code for toggling because atmega328p has 3 timers

and every timer has 2 output pins ex: timer2 has OC2A pin 11 and OC2B pin3 and in isr routine u can do other thing.

KevinK91 (/member/KevinK91) ▸ hal4512 (/member/hal4512) a year ago Reply

could you please elaborate more on why the toggle code isn't necessary in this sketch?

hackerh (/member/hackerh) a year ago Reply

so if you wanted an interrupt every second (frequency of 1Hz):

compare match register = [16,000,000 / (prescaler * 1)] -1
with a prescaler of 1024 you get:

metrog (/member/metrog) a year ago Reply

Hi,

Your project and explanations helped me a lot starting with my project. Thanks!

I'm using Arduino uno timer 0 and 2 to output 2 distinct (in sync) MIDI Clock signal and it works like charm. I can change BPM with a knob going from 30 to 250.

I'm struggling on my next step which is introducing an offset (-200ms / +200ms) between the 2 signals using an analogRead on a knob.

Would you know how to induce such an offset for a Serial output?

Timr0 and Timer2 doesnt have much room to change start point and end point and the induced delay would only be a few milliseconds maybe.

Well if this message finds you and you wish to reply, I'm all ears opened.

Thanks again,

Stef

tejonbiker (/member/tejonbiker) a year ago Reply

Wonderful, I worked with PICs but I see the AVR work slighly different

youraagh (/member/youraagh)

Hi, great tutorial, thanks heaps. 2 years ago Reply

What libraries do I need to include, I'm getting compiler errors saying it doesn't know the names of the timing registers.

PravinahS (/member/PravinahS) 2 years ago Reply

Hi Amanda,

I found these article very informative. I'm designing a card reader and i have included these timer interrupts in my programs.I have several confusions to use these timer interrupts on Arduino Leonardo.I have attached pins on pin 3 and pin 2 for D0 and D1 from my Mifare card reader.My card is a Mifare (32bits) type, so when i batch these card,i'm getting the extra bit of 33,so i decided to use timer interrupts just to execute exact 32 bits,but still i'm getting these extra bit.Please help me..Im lost!!

SitthinutK (/member/SitthinutK) 2 years ago Reply

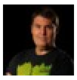
Thanks!!

RosmiR (/member/RosmiR) 2 years ago Reply

In an Arduino code for pulse sensor 256 prescalar is activated by TCCR2b=0x06 . Why is it so ? Why is it not 0x04 as per the table for Timer0. Is it possible to make an interrupt of 100 Hz using Timer 2. (ocr2a <256, 1024 prescalar is not present). Could you help me with this. I am just a beginner.

RosmiR (/member/RosmiR) 2 years ago Reply

How could I make an interrupt of 100Hz with timer 2. Could you share the table for timer2 similar to the one for Timer0 .



ProfRobNewton (/member/ProfRobNewton) 2 years ago Reply

Just wondering if it would not be easier and more accurate to generate an interupt when a switch change is registered and use millis() to determine the time since the last switch change? Sampling every 1ms using the method above may miss a switch change??

andycomic (/member/andycomic) ▸ ProfRobNewton (/member/ProfRobNewton) 2 years ago Reply

I am no expert but to my knowledge the problem with switch state interrupts is that they bounce and your can't denounce a switch using software on an interrupt routine as its already been triggered (as in the interrupt routine is already done before you have a chance to de-bounce.). The other way is to use a hardware based button debounce switch which just uses a capacitor. This will solve the problem. However unless your project is huge, polling a switch every x cycles doesn't really make any difference to any project.

Fardenco (/member/Fardenco) ▸ andycomic (/member/andycomic) Reply 2 years ago

I had the problem with a motor encoder, just put a 10k Ohms resistor to Vcc and a 1µF capacitor to GND around your interrupt pin, this will be ok !!
Pin interrupt would be more efficient because you'll never miss a change, and above that, you'll not waste time with useless interruption =)

DavidS13 (/member/DavidS13) ▸ ProfRobNewton (/member/ProfRobNewton)

2 years ago Reply

While I don't think sampling every 1ms may miss a switch change (think about it, 1000 times a second???) I too would like to know if the more efficient way would be to make an interrupt for the switch change. I have a project that would use something similar and I'm trying to determine the most efficient course of action.

---

manoch_dc (/member/manoch_dc) 2 years ago Reply

thank for Example interrupts that working.

!!But void loop() has not work

my increase program

void setup(){

//set pins as outputs

pinMode(7, OUTPUT); //my led

pinMode(8, OUTPUT);

pinMode(9, OUTPUT);

pinMode(13, OUTPUT);

cli();//stop interrupts

same example

sei();//allow interrupts

}

void loop(){

digitalWrite(7, HIGH); // turn the LED on (HIGH is the voltage level)

delay(100); // wait for a second

digitalWrite(7, LOW); // turn the LED off by making the voltage LOW

delay(100); // wait for a second

//do other things here

}

---

coltonschlosser (/member/coltonschlosser) 2 years ago Reply

I made a python script that walks you through setting up the code for timer interrupts.
https://gist.github.com/cltnschlosser/e747efae07a76e117e02

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killercatfish (/member/killercatfish) 2 years ago Reply

Could this be used for a 2 player reflex timer game? Trying to push your players button on multiples of 5 seconds and getting a point when you do.

---

zinedine (/member/zinedine) 2 years ago Reply

Good job, I want to import 200khz frequency, it's about 5us, but I only have 13.6us with every OCRnA that I used to change from 0 to 80. I'm using 2560. How can I do that?

Thanks

---

faizal91 (/member/faizal91) 2 years ago Reply

Thank you... Very nice tutorial!

EnekoL (/member/EnekoL) 2 years ago Reply

Hei, very usefull application, but i´ve one cuestion, how can i call de interrup for timer 1 inside the loop??Thanks!

ohoilett (/member/ohoilett) ▸ EnekoL (/member/EnekoL) 2 years ago Reply

I could be wrong. You may want to check

http://arduino.cc/en/Reference/AttachInterrupt

ohoilett (/member/ohoilett) ▸ EnekoL (/member/EnekoL) 2 years ago Reply

You don't call interrupts like normal functions. An interrupt service routine (ISR) is "called" or better yet "triggered" at the specified frequency that you determined when you set up the interrupt.

thegoodhen (/member/thegoodhen) 3 years ago Reply

Hello... The code is very useful, however, be aware that the sketch you posted doesn't seem to ever get into the main loop!

DavidS13 (/member/DavidS13) ▸ thegoodhen (/member/thegoodhen) Reply 2 years ago

I'm kind of a noob about all this, but unless I'm mistaken the point of interupts is that you don't NEED a main loop. The main loop just runs forever and everything is done when interupts happen. It is an extremely efficient way to code....

Unidentified Identity (/member/Unidentified Identity) 3 years ago Reply

can we generate square wave with varying pulse width with this interrupt ?

amandaghassaei (/member/amandaghassaei) (author) ▸ Unidentified Identity (/member/Unidentified Identity) 3 years ago Reply

yes it's possible, but probably easier to just use the analogWrite command

srinivas_arduino (/member/srinivas_arduino) 3 years ago Reply

sir please send program for generating 15000 sample valuesin 10 sec using timer interrupt values adc values from accelometer sensor

srinivas_arduino (/member/srinivas_arduino) 3 years ago Reply

sir please send program for generating 15000 sample values using timer interrupt values adc values from accelometer sensor

AbdulW87 (/member/AbdulW87) 3 years ago Reply

i am confused about something in the code, the line

// turn on CTC mode

TCCR0A |= (1 << WGM01);

shouldnt it be TCCR0A |= (1<<WGM02); or am i understanding this wrong, i am a beginner at arduino so its still very confusing to me.

O-Zo (/member/O-Zo) 3 years ago Reply

Thank you for this tutorial, I'm struggling with a project and this helped me stumble at least to the right direction.

More Comments